

CIV102 Project Team 509 Design Report

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I. Introduction

The purpose of this CIV102 bridge assignment is to design and construct a stable and efficient structure capable of safely carrying applied loads across its span while meeting all project constraints. The goal is to demonstrate an understanding of structural behavior, load paths, and practical engineering decision-making. By integrating concepts of failure modes, load distribution, and geometric optimization, the final design aims to maintain structural integrity under bridge load cases 1 and 2, minimize unnecessary weight, maximize strength-to-weight ratio, and illustrate the fundamentals of effective bridge design. With these fundamentals in mind, our bridge design aims to achieve these goals through an iterative approach shown in our process work.

II. Background

Overall Bridge Description

The final bridge is a revised beam-bridge design measuring 1256 mm in length, built using a thin-walled box-girder configuration. Its primary features include a double-layered top deck spanning 120 mm, a continuous internal spine extending 360 mm, and 16 diaphragms arranged along the length of the structure to provide internal stability (see **Figure 1**).

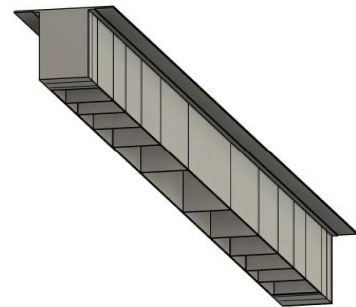


Figure 1: 3D Model of the final layout of the bridge, showcasing the general shape, diaphragms, and cross-section.

Top Deck

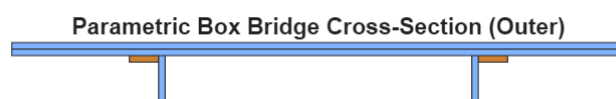


Figure 2: Top Deck of Bridge Cross-section, made by overlapping 2

The top deck made with two layers of matboard, each 1.27 mm thick and approximately 120 mm long (see **Figure 2**). Laminating these sheets increases flexural compressive capacity and improves overall deck stiffness. The two-layer system also enables the inclusion of deck splices while maintaining strength and reducing the likelihood of localized failure.

Walls

The vertical walls of the box girder were designed with a height of 117.5 mm to increase the moment of inertia along the span, particularly near mid-span where bending moments are highest (see **Figure 3**). This increased height also raises the neutral axis, improving resistance to flexural compression.

Bottom

The bottom of the bridge remains open along most of its span, except for closed support regions measuring 60 mm at each end. This design choice raises the neutral axis, enhancing flexural compressive strength while reducing unnecessary material usage.

Spine

A continuous spine was added along the top interior of the box cross-section to provide additional stiffness and support at mid-span where the bending moment reaches its maximum (see **Figure 4**). The spine reinforces the deck against compressive buckling and contributes to overall stability.

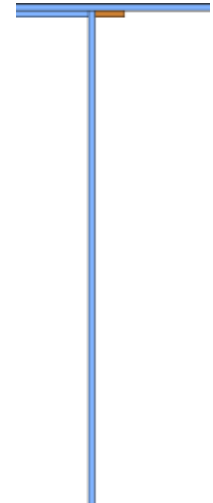


Figure 3: A wall of the cross section, showing its 120 mm height and glue tab.

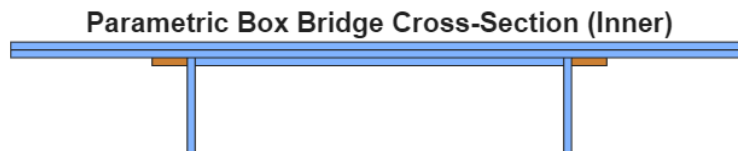


Figure 4: The top deck with the spine included, centered in the bridge to support against flexural buckling.

Diaphragms

The structure incorporates two types of diaphragms: inner and outer, arranged in a quadratic spacing pattern along the span (see **Figure 5**). These diaphragms prevent local wall deformation, assist in distributing loads more evenly, and strengthen the box girder against torsional effects.

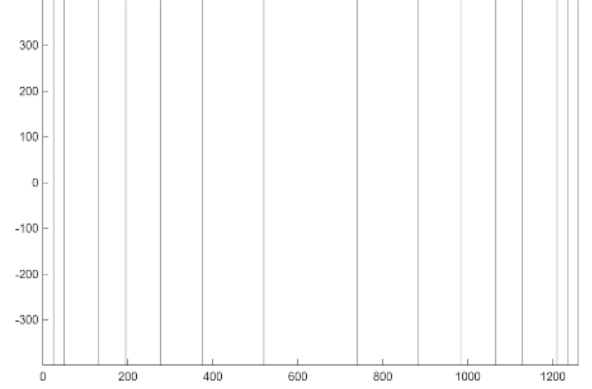


Figure 5: MATLAB code output showcasing the spacing of the 16 diaphragms used in the bridge, concentrated at the ends.

Load Boxes

Load boxes were incorporated at both ends of the structure to ensure consistent and stable load application during testing. These boxes interface with the deck and walls to distribute the applied load in the areas where shear force is the highest. As seen in **Figure 5**, the diaphragms concentrated at the ends act as these load boxes to prevent the chance of shear buckling.

Throughout the iterative process, various equations are used for both hand calculations and the MATLAB code to ensure our cross-section is optimized in both its properties and efficiency. To calculate the neutral axis, the equation

$$\bar{y} = \frac{1}{A_{total}} \sum A_i d_i \quad (1)$$

is used, $\bar{y}$ is the location of the neutral axis from bottom in mm, A_{total} is the total area in mm^2 of the cross-section, and $\sum A_i d_i$ represents the sum of each individual subarea multiplied by its distance from the subarea's centroidal axis to the bottom, expressed in mm^3 . To calculate the second moment of area, the equation

$$I = \sum (I_{oi} + A_i d_i^2) \quad (2)$$

is used, in which I_{oi} is in mm^4 , and $A_i d_i^2$ is in mm^4 . The equations,

$$\sigma_{tens} = \frac{M y_{bot}}{I} \quad (3)$$

$$\sigma_{comp} = \frac{M y_{top}}{I} \quad (4)$$

are used to calculate the flexural stress and compression of the cross-section, where σ is expressed in MPa, M is in N*mm, y is in mm, and I , which represents the second moment of area, is expressed in mm⁴. Shear stress is calculated using the equation

$$\tau = \frac{VQ}{Ib} \quad (5)$$

where V is the internal shear force in N, Q is the first moment of area of the section calculated in mm³, I is the cross-sectional property for the second moment of area, and b is the width of the section where the stress is being calculated.

Local buckling cases are determined using the following equations:

Case 1:

$$\sigma_{crit} = \frac{4\pi^2 E}{12(1-\mu^2)} \left(\frac{t}{b}\right)^2 \quad (6)$$

Case 1 models local buckling in a flat plate that is fully supported along all four edges and loaded in uniform compression. Because the plate is well-restrained, it has a relatively high buckling capacity. The critical stress depends mainly on stiffness (through E) and slenderness (t/b), giving a larger coefficient of 4 in the equation.

Case 2:

$$\sigma_{crit} = \frac{0.425\pi^2 E}{12(1-\mu^2)} \left(\frac{t}{b}\right)^2 \quad (7)$$

Case 2 represents a plate that is compressed but has less restraint along its edges compared to Case 1. With weaker support conditions, the plate buckles more easily. This is why the coefficient drops to 0.425, resulting in a lower critical buckling stress. This case is used when one or more edges of the wall cannot transfer significant compression.

Case 3:

$$\sigma_{crit} = \frac{6\pi^2 E}{12(1-\mu^2)} \left(\frac{t}{b}\right)^2 \quad (8)$$

Case 3 describes a plate that is compressed but experiences stronger support or boundary restraint than Case 1. This increases its ability to resist buckling, producing a higher coefficient of 6. This case applies when the element behaves as a shorter or more rigid plate segment due to stiffening features.

Failure Mode	Critical Load (N)	FOS
Tensile failure	1619	1.62
Compressive failure	385	0.390

Failure Mode	Critical Load (N)	FOS
Shear (matboard)	1007	1.010
Glue shear (top)	3542	3.54
Glue shear (bottom)	1805	1.800
Local buckling (case 1)	933.4	0.930
Local buckling (case 2)	6032	6.03
Local buckling (case 3)	1754	1.75
Local buckling (case 4)	919.8	0.920

Design 0 served as the initial conceptual cross-section developed directly from the assignment. As seen in Figure 1, It consists of several small rectangular elements of uniform thickness arranged into a highly segmented geometry. Using the centroidal equation from Equation 1, the neutral axis is calculated to be approximately 41.4 mm from the bottom, while the moment of inertia computed through Equation 2, yielded $418 \times 10^3 \text{ mm}^4$. Although these results provide a useful baseline, they highlight the inefficiencies of the complex layout.

Structurally, the design places too much material close to the neutral axis rather than at the flanges where it most effectively contributes to bending stiffness. Since bending resistance depends strongly on the square of the distance from the neutral axis, this design iteration would experience higher bending stresses and earlier compressive or tensile failure compared with more efficient shapes like I-beams or box sections. The numerous small rectangles also create many glue seams, which introduce large shear stresses due to narrow bonding widths. These seams become critical shear-transfer regions and reduce the reliability of the section.

Although global buckling dominated failure for this design (rendering detailed local buckling calculations unnecessary), the small component widths also gave poor t/b ratios, indicating that local buckling would occur early if the member were laterally restrained. The construction complexity, excess glueing, and comparatively low bending stiffness make the design heavy and impractical relative to its structural performance. These limitations motivated the transition towards a modified Design 0, to shift more area into the flanges and significantly increase the value of the second moment of area, I .

2. Design 0 Modified

Data:

Height: 77.5 mm

Width: 100 mm

Leg Spacing: 77.5 mm

of Diaphragms: undecided

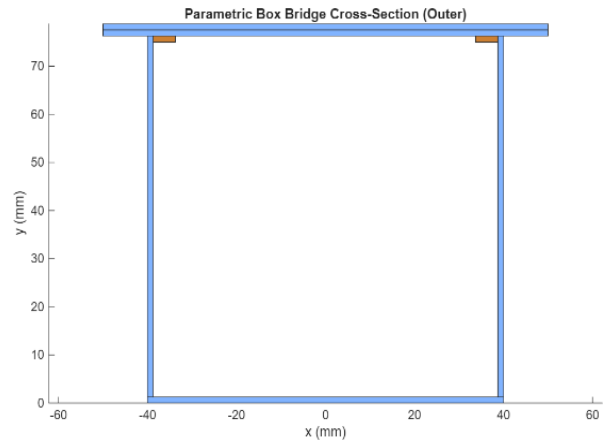
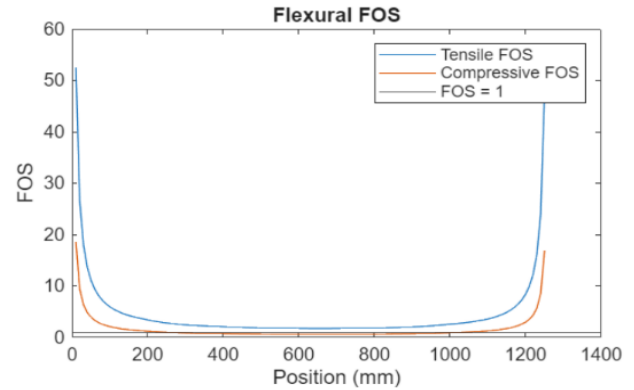


Figure 74: Cross-section of the second iteration, modified to maximize crucial properties, such as the value of the second moment of area and the neutral axis. Design 0 Modified solidified the elimination of a design akin to Design 0 due to its vulnerability to compressive forces.

	Section
Neutral Axis (mm)	49.5
Moment of Inertia (mm ⁴)	0.542x10 ⁶

Failure Mode	Critical Load (N)	FOS
Tensile failure	1753	1.75
Compressive failure	620	0.620
Shear (matboard)	1018	1.020
Glue shear (top)	5733	5.73
Glue shear (bottom)	2935	2.93
Local buckling (case 1)	1504	1.500
Local buckling (case 2)	9717	9.72
Local buckling (case 3)	4374	4.37
Local buckling (case 4)	930	0.930

The next iteration, as seen in Figure 2, was developed as a refined version of Design 0, created by adjusting the rectangular components while maintaining the overall cross-sectional concept. The key intention was to improve bending performance by increasing the top thickness via a double-layered flange, shortening the base, and increasing the height to shift the neutral axis upward. These changes were meant to make the section more efficient in bending without altering too much of the original dimensions. The resulting section produced a moment of inertia of $6.12 \times 10^5 \text{ mm}^4$, determined from Equation 2 and a neutral axis calculated through Equation 1 located 57.3 mm from the bottom, indicating a measurable improvement from the original Design 0 properties.



--- FLEXURAL STRESS AND FOS (Matboard) ---
 Flexural Tension Failure Load: 1753.386 N
 Flexural Compression: 620.453 N
 Minimum Tensile FOS: 1.753
 Minimum Compressive FOS: 0.620

Figure 8: A graph of the FOS for flexural compressive failure along the bridge. FOS = 1 is also graphed as a benchmark. Note that the graph reaches a maxima at the supports and a minima roughly at the center.

Despite the improvement in stiffness, the governing failure mode for this section is compressive stress, occurring at a predicted load of 620 N as seen in Figure 3. Since bending produces maximum compressive stresses at the top surface and the compression flange in this design remains relatively slender, the section becomes highly vulnerable to buckling or crushing in compression. This is reinforced by the compressive stress equation from Equation x, which shows that even with a higher moment of inertia, insufficient flange thickness or width limits compressive strength. The local buckling calculations further demonstrate this imbalance: some place exhibit extremely high σ_{crit} values such as case 2 where $\sigma_{crit} = 9.12 \times 10^3 \text{ N}$, calculated through Equation 7) while others are comparatively weak, such as case 4 with a σ_{crit} value of $0.930 \times 10^3 \text{ N}$ (calculated with Equation 9). Some elements are over-designed while others remain governing failure planes.

The shear results also highlight inefficiencies. Although shear strength in the matboard and glue shear strength are not critical modes, the reliance on multiple glued interfaces reduces reliability. Combined with the complexity of this layout, the design introduces unnecessarily weak areas and increases the likelihood of construction errors, which would further reduce the effectiveness of the improved second moment of area and worsen buckling performance.

These shortcomings helped us determine possible next steps that must be taken:

1. Increase compression-flange thickness or reduce its unsupported width to raise local buckling capacity
2. Change into Standard I-Beam
 - High efficiency (two flanges far from neutral axis)

- A web to resist shear force
 - Material most likely minimized while achieving high value for second moment of inertia
3. Box Beam/Closed Section
- Resistance to local buckling
 - Light, but strong
 - Fewer glue seams => higher reliability

3. I-Cross Section

This design iteration explored the use of an I-shaped cross-section because of its favorable characteristics of placing most material far from the neutral axis, which theoretically increases bending stiffness. As shown in Figure 9, the section uses a tall, narrow web with thin flanges, resulting in a total area of 432 mm^2 , a centroid located 71.3 mm from the base, and a significantly larger moment of inertia of $1.558 \times 10^6 \text{ mm}^4$ compared to prior iterations. This substantial increase in I directly translates to higher flexural stiffness and reduces mid-span deflection under equivalent bending moments. In principle, such geometry should offer excellent bending efficiency.

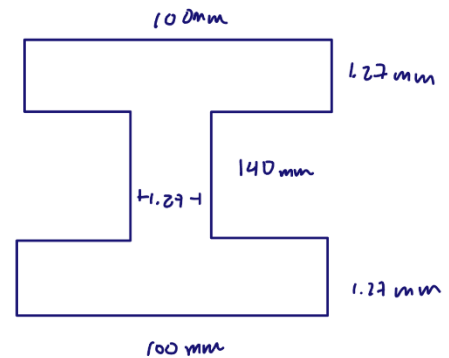


Figure 9: Potential dimensions of an I-beam cross-section. Due to extreme vulnerability to local buckling, this design is discarded.

However, the cross-section's proportions introduce critical weaknesses. The web's thickness-to-height ratio is extremely small, which makes the web susceptible to local buckling well before the member reaches its theoretical strength. Once local web buckling begins, the stiffness and load-carrying capacity collapse rapidly, a risk that must be unacceptable. Similarly, the flanges are only 1.27 mm thick, rendering them too slender to resist compressive loads or concentrated forces applied at the deck surface. These thin flanges are likely to deform, fold or even separate from the web at the glue seams.

Additionally, the I cross section suffers from torsion in cases of eccentric loads, which are loads that are not applied on the shear axis of a cross section. The shear axis for an I cross section is at its vertical or horizontal lines of symmetry. So, any loads applied asymmetrically on the top side of this cross section will cause torsion. Considering human error that the train may not be perfectly centered due to the lack of tracks, there is risk for torsion, which would thus exacerbate any shear effects on the bridge [1].

Considering these weaknesses prompted a select few key ideas that may be implemented for the next iteration:

1. If keeping the I cross-section, double the flange and add diaphragms to reduce local buckling possibilities
2. Convert to a hollow squared/rectangular section by folding the flanges down and glueing it to the web
 - May help with local buckling
3. Consider a different cross-section, such as a rectangular cross-section that is short on the base and tall to increase the value of neutral axis and increase area at same time (increases moment of inertia value)

4. Split I-Cross Section

Data:

Height: 123.8 mm

Width: 100 mm

Leg Spacing: 5 mm

of Diaphragms: undecided

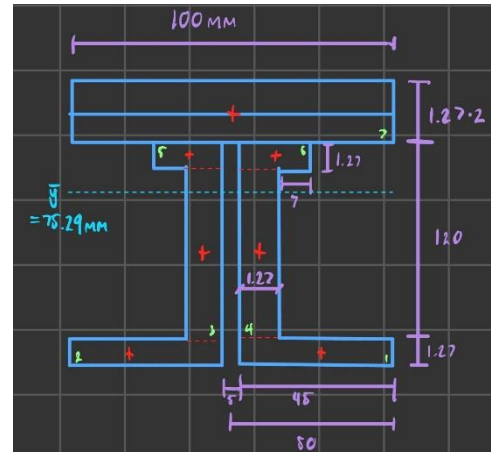


Figure 10: Takes inspiration from a traditional I-Beam but splits it at the leg, however resulted in too much wasted material for not enough strength, thus discarding this cross section

Neutral Axis (mm)	75.3
Moment of Inertia (mm ⁴)	1.667 x 10 ⁶
Area (mm ²)	691
y _{bottom} /y _{top} (centroidal axis top vs. bottom ratio)	1.68

Failure Mode	Critical Load (N)
Tensile failure	991
Compressive failure	334
Shear (matboard)	764

The fourth iteration, as seen in **Figure 10**, takes inspiration from an I-beam, splitting it down the center leg to create a more box-like design, while keeping the benefits of an I-beam. This is done to try and improve the torsion effects as highlighted for regular I-beams in design iteration 3. Additionally, more material towards the top and second leg, raises the centroidal axis and reduces compressive effects on a beam like this, though it seems to be insignificant since the compressive failure load is less than that of design 0. Therefore, this shows how inefficient this

design is with its increased cross-sectional area, yet weaker compressive failure. Additional inefficiencies in this design are shown by the large disparity between the compressive failure loads compared to the other failure loads being more than 2 to 3 times larger. Finally, the ratio between the distance to the centroidal axis from the top versus from the bottom being only 1.68 when the ratio should ideally be as close to 5 as possible shows how inefficient this design is.

Overall, due to the lack of improvement with this cross section, especially being too weak under compression, this design is not iterated further. I-beam shaped designs seem to be far too impractical to consider because of torsion and the complexities involved with fully calculating and considering how torsion will impact the failure load of the bridge, particularly with shear. Thus, further designs will take inspiration from more box-like designs such as HSS or rectangular cross sections.

5. U-Shaped Legs and Triple Layered Top

Data:

Height: 120mm

Width: 140 mm

Leg Spacing: 80mm

of Diaphragms: 18

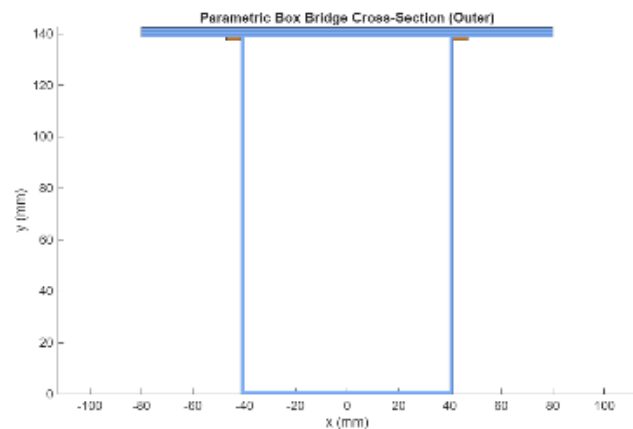


Figure 11: U-shaped leg cross-section with a triple-layer deck. This section consists of tall U-shaped vertical legs supporting a thickened three-layer deck. The bottom plate reduces the neutral-axis height and concentrates shear along the glue seams.

Section	Outer Section	Inner Section (with Diaphragm)
Neutral Axis (mm)	87.6	89.4
Moment of Inertia (mm ⁴)	1.914x10 ⁶	1.963x10 ⁶

Failure Mode	Critical Load (N)	FOS
Tensile Failure	3520	3.52
Compressive Failure	1995	1.990
Shear (Matboard)	1532	1.530
Glue Shear — Top	15 450	16.45

Failure Mode	Critical Load (N)	FOS
Glue Shear — Bottom	8390	8.39
Local Buckling — Case 1	4320	4.32
Local Buckling — Case 2	3780	3.79
Local Buckling — Case 3	10510	10.51
Local Buckling — Case 4	810	0.820

The modified U-shaped leg configuration with a triple-layered deck shown in **Figure 11** represents a refinement of the original cross-section by removing unnecessary interior rectangles and creating a cleaner, more structurally efficient geometry. This iteration aims to increase the bending stiffness through a thickened top flange, simplify the internal layout, and reduce the likelihood of global instability compared to Design 0 and its modified version, and previous I-shaped attempts.

The resulting section, however, produces several notable mechanical consequences. First, the addition of the bottom matboard layer that partially encloses the U-shape drives the neutral axis downward, as governed by the centroidal formula from Equation 1. This is detrimental because the top flange remains both thin and wide, giving it a poor t/b ratio and rendering it highly susceptible to local buckling as predicted by the relevant ratios, such as Equation 6 for case 1 buckling conditions. This increased compressive demand combined with weak buckling resistance creates a premature failure risk long before global bending limits are reached.

In addition, the shear analysis indicates that the adhesive joints between the deck and the vertical legs must carry disproportionately large shear forces, as seen from Equation 5. As Q is large for this cross-section and the glued contact length b is relatively short, the glue seams experience significantly higher stress than the matboard itself. This makes the design materially inefficient, since glue-dominated shear transfer contributes little to stiffness or buckling resistance and introduces a point of vulnerability does not present in more integrated and complex geometries.

Moreover, the calculated material area required to build this configuration exceeds the available matboard allotment, making the full construction infeasible even before cross-sectional property considerations. This alone eliminates the design from practical consideration, but the iteration remains valuable as a conceptual turning point in the design process. It highlights how rectangular or box-influenced geometries offer advantages in stiffness, shear distribution, and buckling resistance, and demonstrates that careful control of the neutral axis location and flange slenderness is essential to achieving a high-efficient bridge design.

From a structural standpoint, the key issues motivating rejection of this design and a need for more iterations include the downward-shifted neutral axis, which reduces the effective thickness of the compression flange and lowers the overall bending capacity; the over-reliance on glue shear transfer, which is potentially failure-inducing; and the high likelihood of local buckling in the top flange due to its wide unsupported width. These factors outweigh the benefits introduced. Thus, the most promising strategies moving forward involve eliminating the bottom plate and reallocating that material toward stiffeners or increased section height and converting the U-legs into a more stable box-shaped form, adding vertical stiffeners to reduce unsupported decks and increase local buckling strength.

6. Box Shaped Legs

Data:

Height: 200mm

Width: 100mm

Leg Spacing: 80mm

of Diaphragms: 18

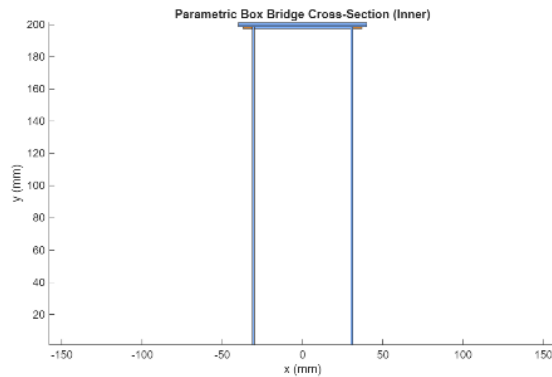


Figure 12: Double layer deck with long U-shaped legs. This iteration was very tall, making the I extremely high and increasing the flexural compression FOS.

Section	Outer Section	Inner Section (with Diaphragm)
Neutral Axis (mm)	133.8	138.4
Moment of Inertia (mm ⁴)	3.39×10^6	3.61×10^6

Failure Mode	Critical Load (N)	FOS
Tensile failure	4185	4.18
Compressive failure	1728	1.730
Shear (matboard)	2217	2.22
Glue shear (top)	1.975×10^4	19.75
Glue shear (bottom)	9972	9.97
Local buckling (case 1)	3739	3.74

Failure Mode	Critical Load (N)	FOS
Local buckling (case 2)	3.551×10^4	35.51
Local buckling (case 3)	2182	2.18
Local buckling (case 4)	804	0.800

The box shaped legs iteration had a massive increase in height and an addition top layer as we noticed the main problem was consistently compressive failure. Along with this, having a smaller top allowed us to avoid local buckling in a case with 3 fixed ends. However, this design came at a great cost to the efficiency of the design.

The large walls increased our changes of shear buckling as well as the area of our diaphragms. Because of this, our failure case was consistently local shear buckling until we added an absurd number of diaphragms, far more than our matboard could fit.

There was also a sufficient lack of efficiency in our design. Many factors of safety, including for local buckling case 2, far exceeded the 1-3 range we planned, meaning that our overall design turned out to be more unrealistic when translated to construction.

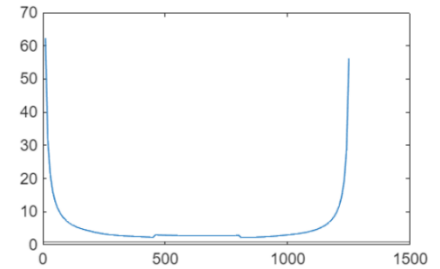
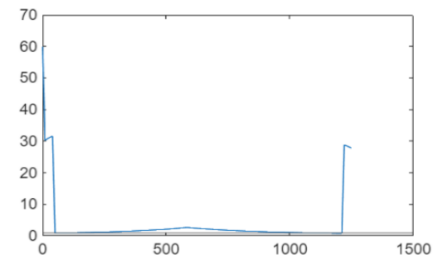


Figure 13: Graph of the FOS for case 2 local buckling as along the bridge. FOS = 1 is also graphed as a benchmark. Note the minima around the center and the maxima around the supports. The spine also provides noticeable support, ‘lifting up’ the lowest part of the FOS graph



--- LOCAL BUCKLING CASE 4 (Shear buckling of leg webs)

Case 4 Local Buckling Failure Load: 804.252 N

Minimum Case 4 Local Buckling Failure FOS: 0.804

Figure 14: Graph of the FOS for case 4 local buckling as along the bridge for this iteration. Figure 15: Graph of the FOS for case 4 local buckling as along the bridge for this iteration. FOS = 1 is also graphed as a benchmark. The FOS graph dips below FOS = 1 around the supports, showing a high likelihood of case 4 shear buckling

7. PI Symbol Double Layered Iteration 1

Data:

Height: 160mm

Width: 140mm

Leg Spacing: 80mm

of Diaphragms: 23

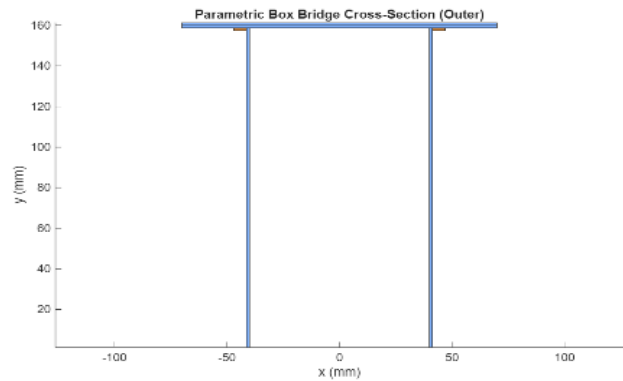


Figure 15: Pi beam with a double-layer deck. This section consists of two tall vertical legs attached to a double-layered deck. The tall shape reduces the flexural compressive stress and helps reduce shear

Section	Outer Section	Inner Section (with Diaphragm)
Neutral Axis (mm)	107.1	109.3
Moment of Inertia (mm ⁴)	1.486x10 ⁶	1.536x10 ⁶

Failure Loads and FOSs

Failure Mode	Critical Load (N)	FOS
Tensile failure	2255	2.26
Compressive failure	1524	1.520
Shear (matboard)	1518	1.520
Glue shear (top)	1.099×10 ⁴	10.99
Glue shear (bottom)	5607	5.61
Local buckling (case 1)	3297	3.30
Local buckling (case 2)	1591	1.590
Local buckling (case 3)	7774	7.77
Local buckling (case 4)	1341	1.340

The pi symbol a good balance between keeping the neutral axis in a strong position and reducing the chance of shear buckling. The structure was planned so that both bending and shear forces are handled effectively.

The results from the calculations show that the chosen cross-section performs well when the whole bridge is under load. The double-layer top deck raised the center of the section and increased its stiffness, which helped prevent local buckling. The bottom layer was removed because it was not needed to carry tension, and this helped save material. The extra material could then be used for diaphragms, which were very important for the overall strength of the bridge.

Overall, the Factors of Safety were balanced. Most of them were between 1 and 3, showing that the bridge used material efficiently while still staying safe. The results for local buckling and compressive failure in the most critical case were also close to each other, which shows that the design did not waste material and worked well as a whole.

The main reason this design was not chosen was due to the inability to fit its dimensions into our matboard. Initially, we assumed that this design could fit into the matboard due to our program. However, we quickly realized that this was only due to us forgetting to put one of our walls into the matboard. Due to this, there was no possibility of achieving this design without removing diaphragms that quickly resulted in an early shear buckling failure.

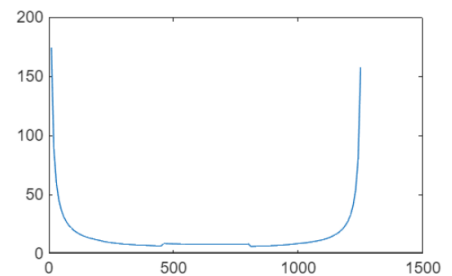


Figure 16: Graph of the FOS for case 3 local buckling as along the bridge. FOS = 1 is also graphed as a benchmark. Note the minima around the center and the maxima around the supports. The spine also provides noticeable support, ‘lifting up’ the lowest part of the FOS graph. It is evident from the graph that the bridge is at very little risk of failure from case 3 local buckling.

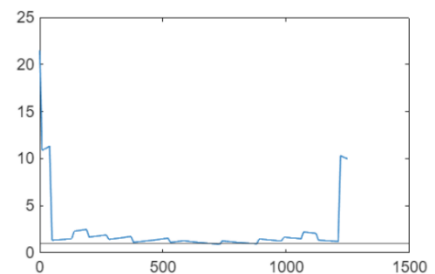


Figure 17: Graph of the FOS for case 4 local buckling as along the bridge. FOS = 1 is also graphed as a benchmark. Note the minima around the center and the maxima around the supports. Each jump in the graph is due to a diaphragm which helps “lift up” the FOS graph, preventing it from falling below FOS = 1.

8. PI Symbol Double Layered Iteration 2

Data:

Height: 120mm

Width: 120mm

Leg Spacing: 60mm

of Diaphragms: 16

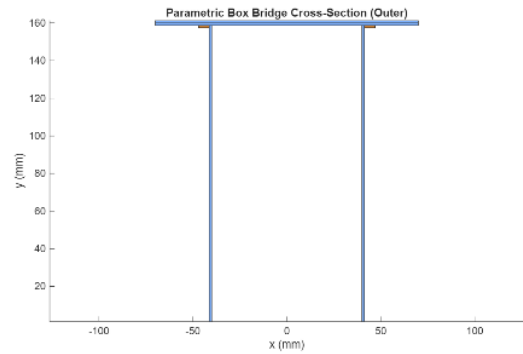


Figure 18: Same cross section as iteration 7 with smaller dimensions to fit the matboard, balancing the two critical fail cases of local buckling case 4 and compressive failure

Section	Outer Section	Inner Section (with Diaphragm)
Neutral Axis (mm)	89.8	93.2
Moment of Inertia (mm ⁴)	0.899×10^6	0.996×10^6

Failure Mode	Critical Load (N)	FOS
Tensile failure	1644	1.64
Compressive failure	1007	1.010
Shear (matboard)	1302	1.31
Glue shear (top)	9698	9.70
Glue shear (bottom)	4956	4.96
Local buckling (case 1)	3793	3.79
Local buckling (case 2)	1909	1.910
Local buckling (case 3)	6119	6.12
Local buckling (case 4)	965	0.960

Our final iteration consisted of the same cross section with changed dimensions to finally fit into the space of our matboard (see **Figure 19**). In creating the cross section, we attempted to balance our two most concerning failure cases, flexural buckling and shear buckling, as close as possible.

As a result, our failure load was optimized to its highest possible value under these load conditions for the given cross section.

This iteration also introduced our quadratic spacing for load buckling, as well as the point where we determined our splices as seen in **Figure 21**. The diaphragms and splices were spaced as to best mitigate failure, concentrated away from the center and not overlayed on top of each other. Overall, this was our most balanced cross-section across the length of the bridge.

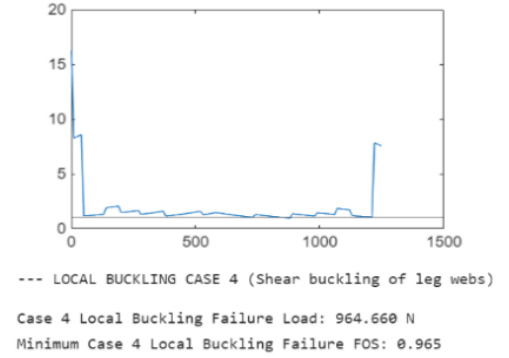


Figure 20: Graph of the FOS for case 4 local buckling as along the bridge. FOS = 1 is also graphed as a benchmark. Note the minima around the center and the maxima around the supports. Each jump in the graph is due to a diaphragm which helps “lift up” the FOS graph, preventing it from falling below FOS = 1

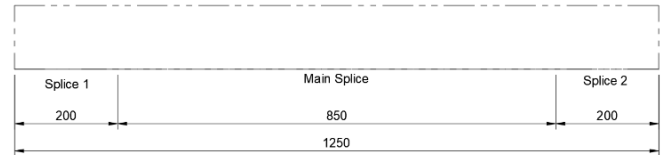


Figure 21: Example Splicing of the wall of the bridge. Both splices lean towards the end, spaced 200mm from the ends, to avoid the concentrate moment in the middle of the cross section.

References

[1] T. Cosgrove and B. Beng, “Design of steel beams in torsion,” 2011. Available: https://www.steelconstruction.info/images/6/6f/Sci_p385.pdf